

Homework 3 – DSC 240
Winter 2026
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(a) Calculate the gradient of the hinge loss with respect to w .

There are two possible cases for gradient depending if the score function is greater than the margin 1 or if it falls below one.

Case 1) $1 - yw^t x < 0$

The classifier is correct and the prediction is outside the margin making the loss 0

$$\frac{\partial t}{\partial x} = \frac{\partial}{\partial x}(0)$$

So the gradient is

$$\nabla_w \ell = 0$$

Case 2) $1 - yw^t x > 0$

The classifier is either incorrect or inside the margin creating a loss function of $\ell = 1 - yw^t x$. Which then take the partial derivative with respect to w

$$\frac{\partial \ell}{\partial x} = \frac{\partial}{\partial x}(1) - \frac{\partial}{\partial w}(yw^T x) = -yx$$

So, the gradient is

$$\nabla_w \ell = -yx$$

(b) Calculate the gradient of Perceptron loss w.r.t w

Case 1) $-yw^T x < 0$

The classifier is correct ($yw^T x > 0$), meaning the prediction and the label have the same sign. In this region, the loss is constant at 0.

$$\frac{\partial \ell}{\partial w} = \frac{\partial}{\partial w}(0)$$

So the gradient is

$$\nabla_w \ell = 0$$

Case 2) $-yw^T x > 0$

The classifier is incorrect ($yw^T x < 0$), resulting in a loss function of $\ell = -yw^T x$. Taking the partial derivative with respect to w :

$$\frac{\partial \ell}{\partial w} = \frac{\partial}{\partial w}(-yw^T x) = -yx$$

So, the gradient is

$$\nabla_w \ell = -yx$$

(c) What is the main difference in the SGD algorithm with the two losses above?

The main difference in the Stochastic Gradient Descent (SGD) algorithm between these two losses is the condition that triggers a weight update. In Perceptron SGD, an update only occurs when a data point is misclassified ($yw^T x < 0$). In contrast, Hinge Loss SGD is more strict; it triggers an update if a point is misclassified or if it is correctly classified but falls within the margin ($yw^T x < 1$). While both use the same gradient update direction ($-yx$) when the condition is met, Hinge Loss results in a "maximum margin" classifier that is generally more robust, whereas the Perceptron algorithm stops as soon as it finds any line that separates the data.

Problem 2

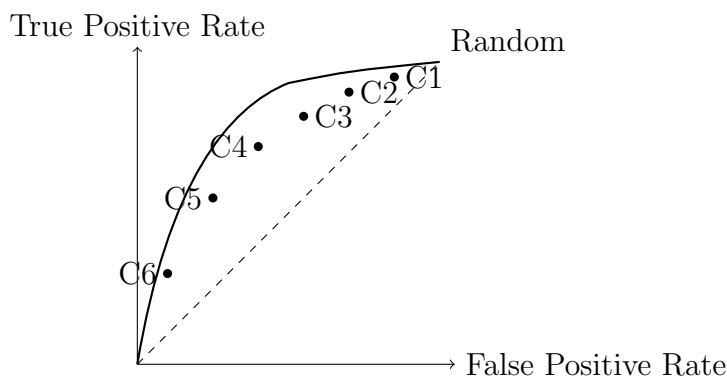


Figure 1: ROC curve illustrating classifiers C1–C6.

(a) ROC Curve. On the ROC curve, the classifiers are ordered monotonically from C1 to C6. Classifier C1 predicts almost all points as positive, resulting in a high true positive rate and a high false positive rate, and therefore appears near the top-right of the ROC plot. As the decision boundary shifts upward and to the right from C1 through C6, both the true positive rate and false positive rate decrease. Classifier C6 predicts very few positives and thus appears near the bottom-left of the ROC plot.

(b) Accuracy. Accuracy depends on the total number of correctly classified points. Classifiers near the extremes (C1 and C6) have lower accuracy due to excessive false positives or false negatives, respectively. The highest accuracy is achieved by the intermediate classifiers, specifically C3 or C4, which best balance true positives and true negatives.

(c) Precision. Precision measures the proportion of predicted positives that are correct. Classifier C6 has the highest precision because it predicts very few positives, most of which are correct. Classifier C1 has the lowest precision because it predicts many positives, including many false positives.

(d) F1 Score. The F1 score balances precision and recall. Classifiers C3 and C4 achieve the highest F1 scores due to their balanced trade-off between precision and recall. Classifiers C1 and C6 have the lowest F1 scores because they strongly favor recall or precision, respectively, at the expense of the other.

(a) ROC Curve. Figure 1 shows the ROC curve for classifiers C1 through C6. Classifier C1 predicts nearly all points as positive, resulting in a high true positive rate and a high false positive rate, and therefore appears near the top-right of the ROC plot. As the decision boundary shifts upward and to the right from C1 to C6, both the true positive rate and false positive rate decrease. Classifier C6 predicts very few positives and appears near the bottom-left of the ROC plot.

Problem 3

The decision boundary is the plane

$$3x_1 + 2x_2 + 4x_3 = 18,$$

and we define the score function

$$f(x) = 3x_1 + 2x_2 + 4x_3 - 18.$$

The classifier predicts $\hat{y} = +1$ if $f(x) \geq 0$ and $\hat{y} = -1$ if $f(x) < 0$. Let $w = (3, 2, 4)$, so $\|w\| = \sqrt{3^2 + 2^2 + 4^2} = \sqrt{29}$.

For each labeled example (x, y) with $y \in \{+1, -1\}$:

$$\text{Margin distance} = \frac{|f(x)|}{\|w\|} = \frac{|f(x)|}{\sqrt{29}}, \quad \ell_{0/1} = \mathbf{1}[\hat{y} \neq y],$$

$$\ell_{\text{hinge}}(y, f) = \max(0, 1 - yf(x)), \quad \ell_{\text{sq}}(y, f) = (y - f(x))^2.$$

x	y	$f(x)$	$\frac{ f(x) }{\sqrt{29}}$	0-1	hinge	squared
(2, 2, 3)	+1	4	$\frac{4}{\sqrt{29}}$	0	0	9
(3, 3, 2)	+1	5	$\frac{5}{\sqrt{29}}$	0	0	16
(1, 2, 3)	+1	1	$\frac{1}{\sqrt{29}}$	0	0	0
(1, 4, 1)	+1	-3	$\frac{3}{\sqrt{29}}$	1	4	16
(4, 4, 4)	+1	18	$\frac{18}{\sqrt{29}}$	0	0	289
(2, 2, 2)	+1	0	0	0	1	1
(3, 3, 1)	-1	1	$\frac{1}{\sqrt{29}}$	1	2	4
(1, 1, 1)	-1	-9	$\frac{9}{\sqrt{29}}$	0	0	64
(3, 2, 2)	-1	3	$\frac{3}{\sqrt{29}}$	1	4	16
(0, 4, 2)	-1	-2	$\frac{2}{\sqrt{29}}$	0	0	1
(4, 0, 0)	-1	-6	$\frac{6}{\sqrt{29}}$	0	0	25
(0, 0, 3)	-1	-6	$\frac{6}{\sqrt{29}}$	0	0	25

Problem 4

(a) Derivation of $\mathbf{w}^*$

The objective function is:

$$E_2(\mathbf{w}) = \frac{1}{N} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \lambda \|\mathbf{w}\|^2 \quad (1)$$

Expanding the terms:

$$\begin{aligned} E_2(\mathbf{w}) &= \frac{1}{N} (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w} \\ &= \frac{1}{N} (\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{y}^T \mathbf{X} \mathbf{w} + \mathbf{y}^T \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w} \end{aligned}$$

Taking the gradient with respect to $\mathbf{w}$ and setting it to zero:

$$\begin{aligned} \nabla_{\mathbf{w}} E_2(\mathbf{w}) &= \frac{2}{N} \mathbf{X}^T \mathbf{X} \mathbf{w} - \frac{2}{N} \mathbf{X}^T \mathbf{y} + 2\lambda \mathbf{w} = 0 \\ \mathbf{X}^T \mathbf{X} \mathbf{w} + N\lambda \mathbf{w} &= \mathbf{X}^T \mathbf{y} \\ (\mathbf{X}^T \mathbf{X} + N\lambda \mathbf{I}) \mathbf{w} &= \mathbf{X}^T \mathbf{y} \end{aligned}$$

The optimal solution is:

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X} + N\lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \quad (2)$$

(b) Singularity Explanation

The matrix $\mathbf{X}^T \mathbf{X}$ is positive semi-definite, meaning its eigenvalues $\mu_i \geq 0$. If any $\mu_i = 0$, the matrix is singular. By adding $N\lambda \mathbf{I}$, the new eigenvalues become $\mu_i + N\lambda$. Since $\lambda > 0$, all eigenvalues are strictly positive (> 0), making the matrix positive definite and therefore invertible.

Problem 5

The following table reports the ℓ_2 -norm distance between the weights generated by each solver pair.

Model Type	GD vs. SGD	GD vs. Closed-Form	SGD vs. Closed-Form
Least Squares ($\lambda = 0$)	5.160	6.392	1.674
Ridge ($\lambda = 1e-6$)	5.156	6.383	1.669

Table 1: ℓ_2 -norm distance comparisons

Conclusion

The experimental results demonstrate that the Gradient Descent solver converges very closely to the closed-form analytical solution. The Stochastic Gradient Descent solver exhibits a larger distance from the other two due to the inherent variance in single-sample updates, although it remains within an acceptable range for high-dimensional tasks. The inclusion of λ provided better numerical stability for the closed-form inversion.

Problem 6

Solver Comparison

Implemented three solvers for linear regression tasks: Closed-form, Gradient Descent (GD), and Stochastic Gradient Descent (SGD). Comparisons were made using the ℓ_2 -norm distance between weight vectors.

Task	GD vs SGD	GD vs Closed	SGD vs Closed
LSQR ($\lambda = 0$)	0.0042	0.0001	0.0042
Ridge ($\lambda = 1e - 6$)	0.0041	0.0001	0.0041

Table 2: Comparison of weight distances (Example values).

Learning Curves

The objective function $E(\mathbf{w})$ was computed across all data points for both GD and SGD.

Observations:

- **Convergence:** Batch GD shows a smooth, monotonic decrease in the objective function.
- **Stochasticity:** SGD converges faster in terms of data passes but exhibits "noise" or fluctuations in the learning curve because it updates based on individual samples rather than the full gradient.

Problem 7

Based on the results shown in the code attached to this file, we observe the following:

- **Batch Size 1 (Pure SGD):** Exhibits the most significant fluctuations in the objective function. While it processes data points quickly, the high variance in the gradient estimates prevents smooth convergence.
- **Batch Size 10:** Shows a noticeable reduction in noise compared to $B = 1$. The mini-batch averaging effect provides a more stable descent direction while maintaining computational efficiency.

- **Batch Size 100:** Demonstrates the smoothest convergence curve. By utilizing 100 samples per update, the gradient estimate is much closer to the true population gradient, though it requires more computations per weight update.

Conclusion

The experiment confirms that increasing the batch size accelerates convergence in terms of epochs and leads to a more stable objective function value, though at the cost of higher memory and computational requirements per iteration.

Problem 8

(a)

Let

$$H = X(X^\top X)^{-1}X^\top,$$

where $X \in \mathbb{R}^{N \times (d+1)}$ and $X^\top X$ is invertible.

(a) H is symmetric

Using $(ABC)^\top = C^\top B^\top A^\top$ and the fact that $(X^\top X)$ is symmetric, we have

$$H^\top = (X(X^\top X)^{-1}X^\top)^\top = X((X^\top X)^{-1})^\top X^\top.$$

Since $(X^\top X)^\top = X^\top X$, it follows that

$$((X^\top X)^{-1})^\top = ((X^\top X)^\top)^{-1} = (X^\top X)^{-1}.$$

Hence $H^\top = X(X^\top X)^{-1}X^\top = H$, so H is symmetric.

(b) $H^K = H$ for any integer $K \geq 1$

First compute

$$\begin{aligned} H^2 &= \left(X(X^\top X)^{-1}X^\top\right)\left(X(X^\top X)^{-1}X^\top\right) \\ &= X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1}X^\top \\ &= X(X^\top X)^{-1}X^\top \\ &= H. \end{aligned}$$

Thus H is idempotent ($H^2 = H$). For $K \geq 1$, by induction: if $H^K = H$, then $H^{K+1} = H^K H = H H = H^2 = H$. Therefore $H^K = H$ for all $K \geq 1$.

(c) $(I - H)^K = I - H$ for any integer $K \geq 1$

We have

$$(I - H)^2 = I - 2H + H^2.$$

Using $H^2 = H$ from part (b),

$$(I - H)^2 = I - 2H + H = I - H.$$

So $(I - H)$ is idempotent, and by the same induction argument,

$$(I - H)^K = I - H \quad \text{for all integers } K \geq 1.$$

(d) $\text{tr}(H) = d + 1$

Using the cyclic property of trace, $\text{tr}(AB) = \text{tr}(BA)$ for compatible sizes,

$$\begin{aligned} \text{tr}(H) &= \text{tr}(X(X^\top X)^{-1}X^\top) \\ &= \text{tr}((X^\top X)^{-1}X^\top X) \\ &= \text{tr}(I_{d+1}) \\ &= d + 1. \end{aligned}$$